Practical 2

Aim: Set up a workspace and create a cluster in Databricks. Verify cluster configuration and execute a simple notebook command.

Objectives: Create workspace, configure cluster, execute notebook.

Outcomes: Students manage clusters and notebooks.

Theory: Databricks is a cloud-based data and analytics platform designed for data engineering, data analysis, machine learning, and artificial intelligence. It provides an integrated environment where users can create notebooks, process large datasets, run SQL queries, develop applications, and perform data visualization.

Databricks provides a workspace that acts as a central environment for organizing notebooks, files, queries, dashboards, and other resources required for data processing.

Input: Databricks workspace.

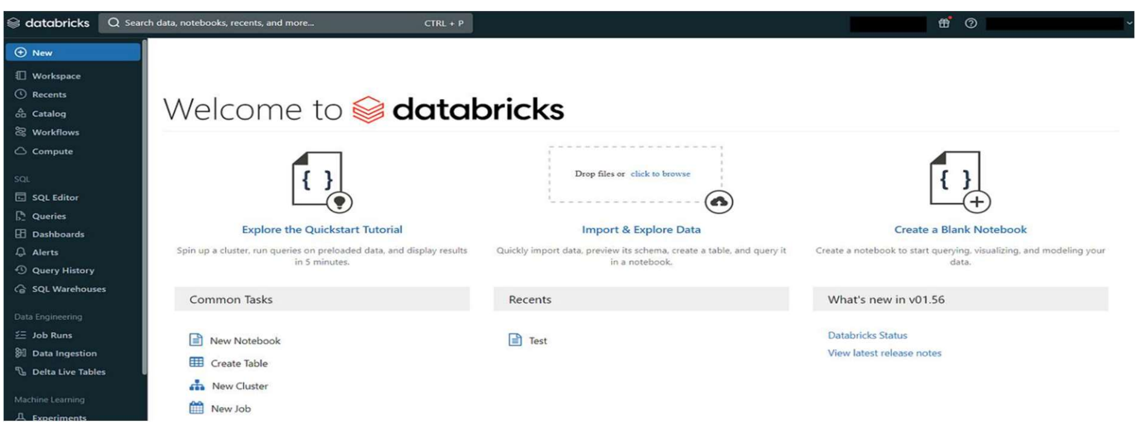
Output:

Step 1: Open Databricks

1. Open https://www.databricks.com/

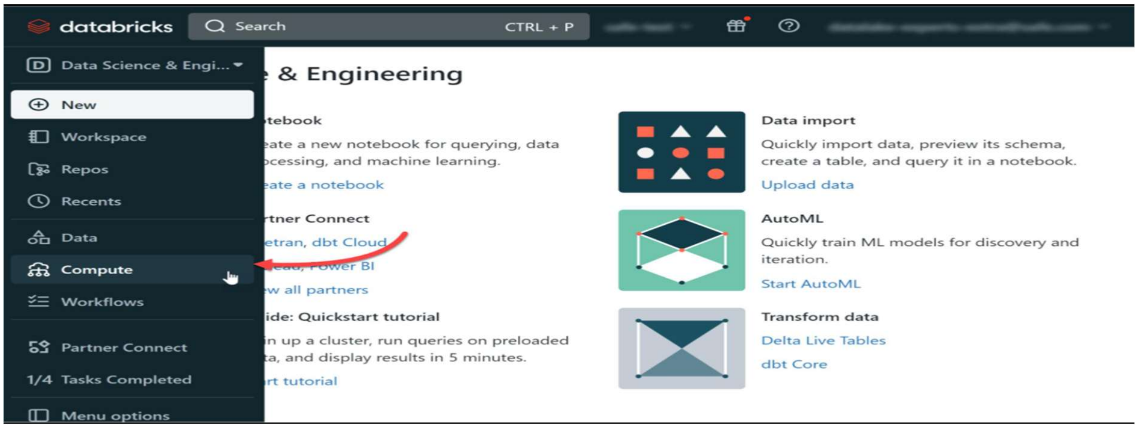
2. Sign in to your **Databricks Free Edition** account.

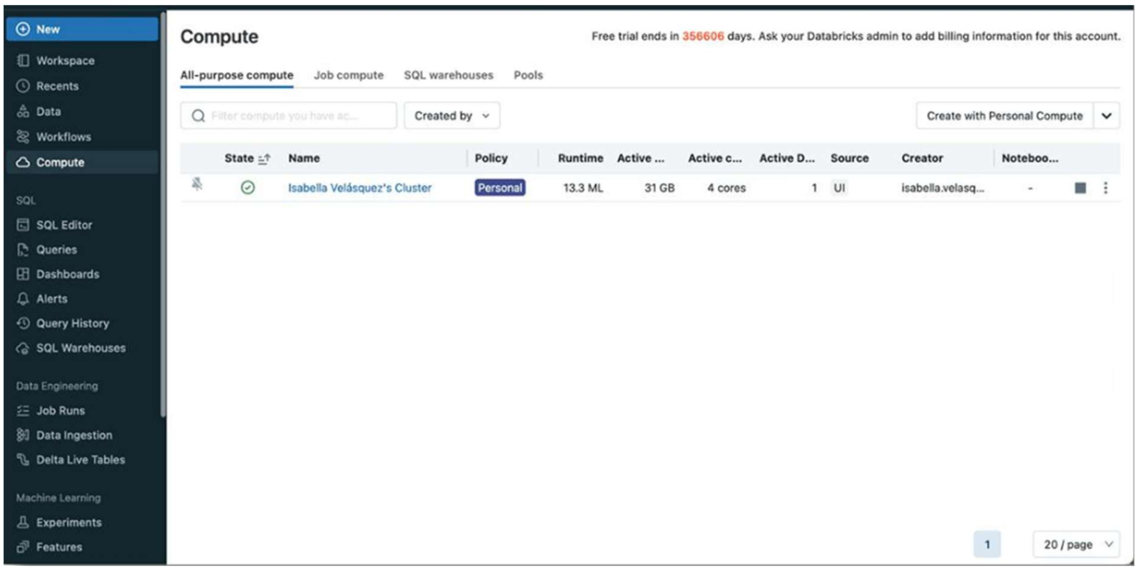
3. After login, you will see the Databricks workspace/home page.



Step 2: Create/Start Compute

1. Select compute to start the cluster.
2. Select the available **Serverless/Compute** option.
3. Start the compute.
4. Wait until the compute becomes ready.





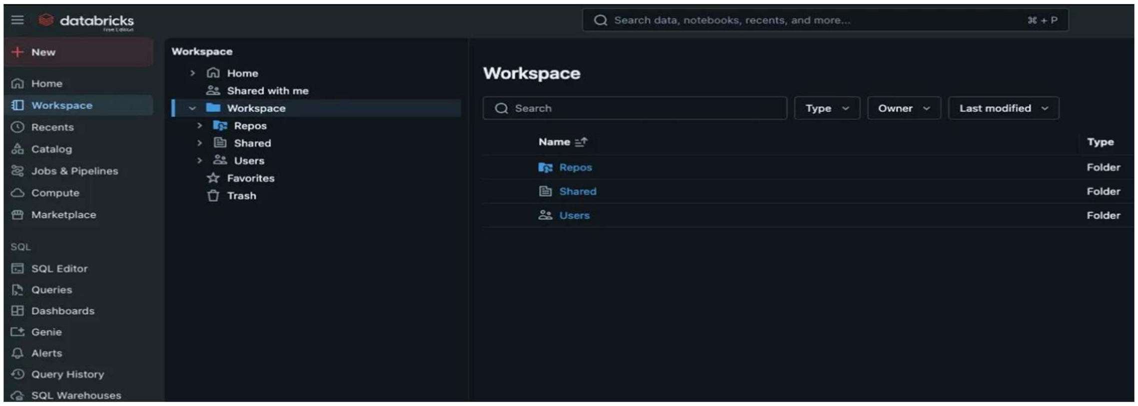
Step 3: Verify Compute Configuration

Open the compute details/configuration available in your Free Edition workspace.

Parameter	Value
Compute type	Serverless / available compute
Status	Running/Ready
Databricks Runtime	As displayed
Runtime language	Python
Compute state	Running

Step 4: Create a Notebook

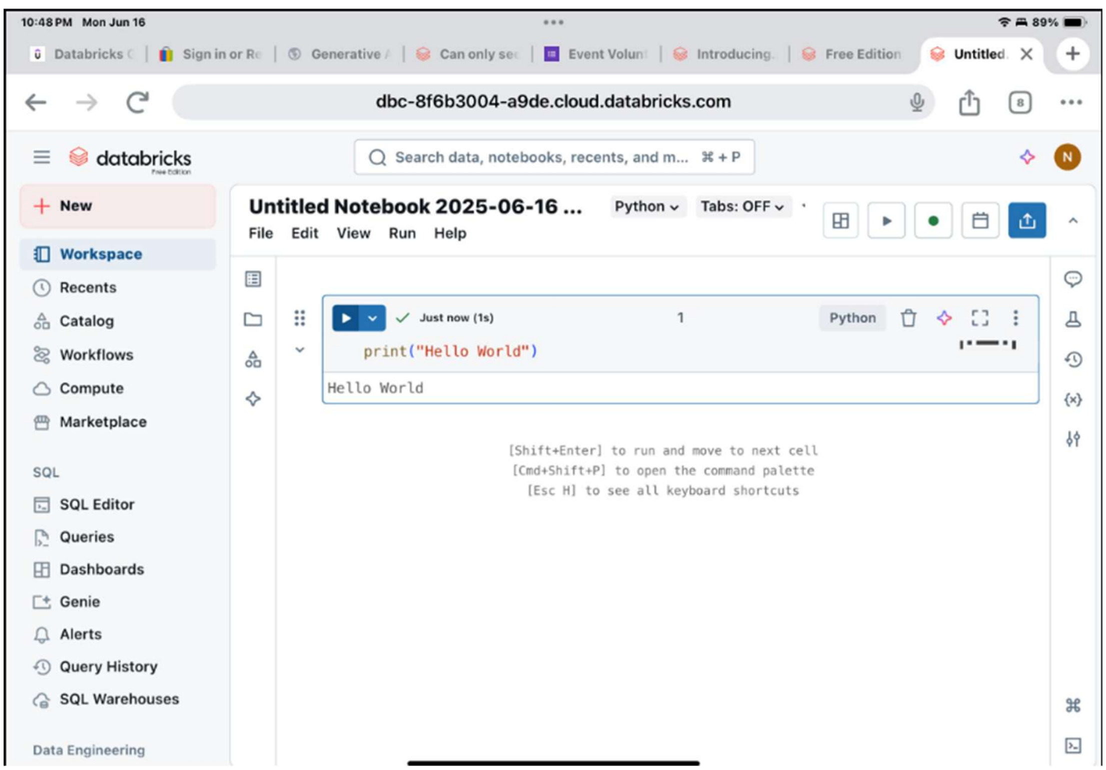
1. From the left navigation, select **Workspace**.



2. Select **Create → Notebook**.

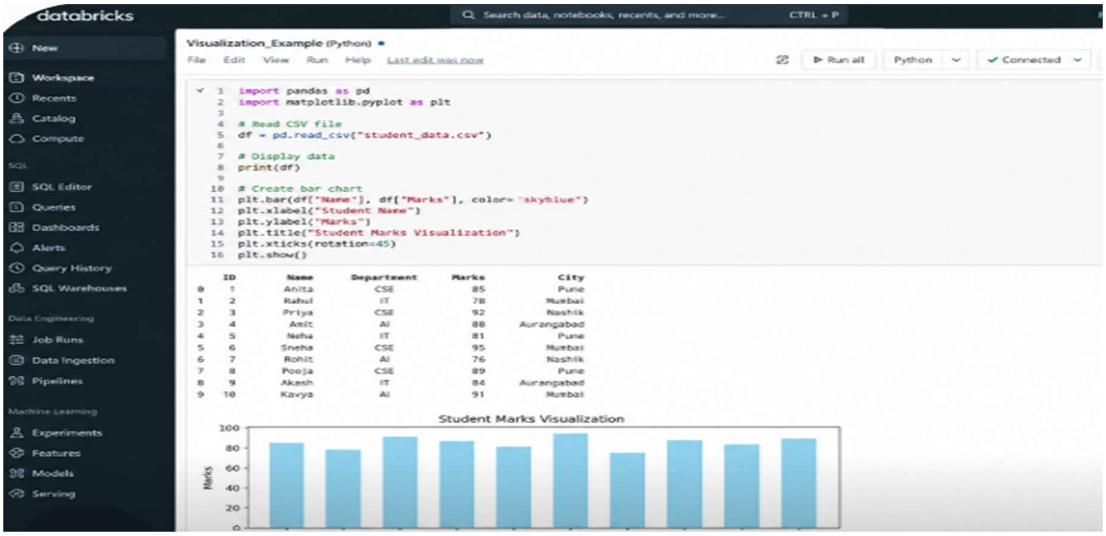
3. Give the notebook a name, for example:
   Practical_2_Databricks

4. Select **Python** as the language.



Step 5: Execute a Simple Notebook Command

In the notebook, enter python program and Click **Run**.



Practical Related Questions:

- What is Databricks Workspace?
- What is purpose of Compute in Databricks?
- Why is it necessary to verify the cluster configuration?
- What is the difference between a workspace and a cluster?

Conclusion: The practical was successfully completed by setting up a Databricks workspace and creating a cluster with the required configuration. The cluster configuration was verified, and a simple command was executed successfully in a Databricks notebook. This practical provided hands-on understanding of the Databricks environment, cluster management, and notebook execution, which are essential for cloud-based data processing and analytics.